

Task:- 8 Normalizing Database using  
6/10/25 functional dependence upto BCNF

Aim:- To normalize database using functional dependent upto BCNF.

Hospital Database.

1. Identify hospital attribute:

Patient — ID, Patient — Name, Doctor — ID,  
Doctor — Name, Department, Room — NO,  
Treatment, Bill — Amount.

2. Define relational schema:

Hospital (Patient — ID, Patient — Name — Doctor — ID,  
Doctor — Name, Department, Room — NO, Treatment,  
Bill — Amount).

3. Determine functional dependence (fDs)  
between attributes.

Patient — ID  $\rightarrow$  Patient — Name, Doctor — ID, Room — NO  
Treatment, Bill — Amount

Doctor — ID  $\rightarrow$  Doctor — Name, Department

Room — NO  $\rightarrow$  Department.

Step 2:- Convert to 1NF

1. Eliminate repeating groups or array
2. Create separate tables for each.  
Repeating group



### Step 3:- Convert to 2NF:-

1. Ensure each non-key attribute depend on the entire primary key.
2. Move non-key attribute to separate tables if key depend only part of the primary key.

- Create Doctor table: Doctor (Doctor — ID  
Doctor — Name, Department).

- Create Patient table: Patient (Patient — ID,  
Patient — Name, Doctor — ID,  
Room — NO, Treatment, Bill — Amount).

### Step 4:- Convert to 3NF

1. Ensure there are no transitive dependence.
2. Move non-key attribute to separate table if they depend on another non-key attribute.

- Create Room table: Room (Room — NO, Department).

• update Doctor table: Doctor (Doctor — ID, Doctor — Name).

### Step 5:- Convert to BCNF

1. Ensure every Department is a candidate key.
2. Check for overlapping candidate key.
3. DeCompose relation to eliminate redundancy.

• No further decomposition needed.



### Using Criteria tool:-

1. Input relation schema and functional dependency.
2. Griffith tool generator as dependency graph.
3. Analyze the graph to identify normalization issues.
4. Apply normalization rules to transform the schema.
5. Verify the resulting schema meets BNF Criteria.

### Griffith tool Steps:-

1. Create a new project in Griffith.
2. Define the relational scheme and try.
3. Run the "Dependency Graph" tool.
4. Analyze the graph for normalization issues.
5. Apply transformations using the "Normaliz" tool.

6. Verify BNF Compliance using the

### "BNF Check" tool

### Normalized Schema:-

1. Patient (Patient-ID, Patient-Name, Doctor-ID, Room-NO, Treatment, Bill-Amount).
2. Doctor (Doctor-ID, Doctor-Name).
3. Room (Room-NO, Department).



| VEL TECH - CSE          |                                                                                       |
|-------------------------|---------------------------------------------------------------------------------------|
| EX NO.                  | 8                                                                                     |
| PERFORMANCE (5)         | 5                                                                                     |
| RESULT AND ANALYSIS (5) | 5                                                                                     |
| VIVA VOCE (5)           | 5                                                                                     |
| RECORD (5)              |                                                                                       |
| TOTAL (20)              | 15                                                                                    |
| SIGN WITH DATE          |  |

Results:-

Thus the Normalized Database using functional dependence upto BCNF executed successfully.